PRIMARY STEM EDUCATION CURRICULUM & PLATFORM

# Eco-Bot Builders

## Game Overview, Rewards & Features

A cross-platform STEM coding game for primary school students (ages 7–11)



**TARGET AUDIENCE**
Primary Students (Ages 7–11)

**STANDARDS ALIGNMENT**
CSTA Level 1B & NGSS Practices

**GAME ENGINE**
Unity 2022.3 LTS (URP) Cross-Platform

**PEDAGOGICAL CORE**
Computational Thinking & Eco-Systems

**Project:** STEM-APS-UniMAP **Initiative:** Universiti Malaysia Perlis (UniMAP) **Release Date:** November 2026

## 1 What Is Eco-Bot Builders?

**Eco-Bot Builders** is a 2.5D isometric puzzle and city-builder game where students program a friendly robot companion — the "Eco-Bot" — using drag-and-drop logic blocks to clean, water, power and rebuild polluted floating islands called "Sky-Isles."

Each level wraps a computational-thinking idea (sequences, loops, conditionals, sensors, subroutines) inside a real-world sustainability problem (recycling, solar, wind and hydro energy, bridges, gardens).

Students write short block programs (Move, Turn, Clean, Water, Repeat loops, If/Else, subroutines, etc.), press Play, and watch the Eco-Bot carry out their code. The goal is to solve each puzzle within a limited "code capacity" (block budget), which rewards efficient thinking — for example, using a Repeat loop instead of many repeated blocks.

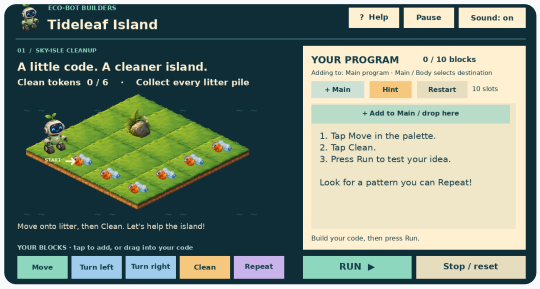



*Figure 1: 2.5D Isometric Sky-Isle cleanup mission with tile grid and solar elements.*

### Code Capacity & Algorithmic Efficiency

Unlike open-ended sandbox environments, each Sky-Isle level enforces an intentional *block capacity limit*. This constraint guides learners to look for repeating movement patterns and algorithmic symmetries, nudging students naturally toward abstraction, loops, and modular programming structures rather than tedious brute-force sequences.

## 2 How To Play (The Core Loop)

The core gameplay loop structures every student session into six clear, iterative milestones:

### 1 Select Level

Pick a level from the Level Select screen. Levels unlock in order as you complete them.

### 2 Read Mission

Read the eco-mission (e.g., "clean all the litter", "clean the dusty solar panels").

### 3 Assemble Code

Drag logic blocks into the program area, staying within the block budget.

### 4 Run & Observe

Press Play to run the program and watch the Eco-Bot carry out your code in real time.

### 5 Adaptive Hints

If stuck, use the 3-tier adaptive hint system to receive scaffolded pedagogical guidance.

### 6 Restore & Unlock

Complete the goal to restore the Sky-Isle, earn a cosmetic reward, and unlock the next level.

## 3 The 10 Levels (Curriculum)

The curriculum advances systematically from basic linear sequences to complex integrated engineering systems, closely aligned with primary school cognitive development curves:

| Level | Title | STEM Concept | Block Budget | Cosmetic Reward |
|---|---|---|---|---|
| 1 | **Sky-Isle Cleanup** | Logic | 10 | **Leaf Antenna** |
| 2 | **Solar Panel Path** | Sequencing & Loops | 6 | **Solar Visor** |
| 3 | **Water Wheel Village** | Nested Loops | 8 | **Water Wheel Wheels** |
| 4 | **Wind Farm Ridge** | Energy Systems | 10 | **Wind Turbine Hat** |
| 5 | **Recycling Plant Sort** | Conditionals | 10 | **Recycled Armor** |
| 6 | **Hydro Dam Hydraulics** | Mathematical Modeling | 12 | **Hydro Jetpack** |
| 7 | **Bridge Builder Canyon** | Structural Engineering | 12 | **Bridge Builder Boots** |
| 8 | **Compost Garden Sensors** | Sensors & Events | 12 | **Compost Cape** |
| 9 | **Night Grid Balancer** | Systems Thinking | 14 | **Aurora Paint** |
| 10 | **Eco-City Master Mission** | Integrated Engineering | 16 | **Golden Gear Crown** |

**Curriculum Implementation Note:** Levels 1 and 2 are currently built and verified as Unity source; levels 3–10 are designed and defined in the curriculum (the full 10-level arc).

## 4 Rewards

Eco-Bot Builders reinforces perseverance and collaborative achievement through multi-tiered reward systems designed specifically for primary students:

### ★ Cosmetic Rewards

Completing a level for the first time unlocks a cosmetic item to customize the Eco-Bot (see the table in Section 3). The final level (10) grants the prestige **"Golden Gear Crown."**

### ⚡ Clean Tokens & Community Progress

Students earn "Clean Tokens" as they play. The whole class works together toward a shared Community goal of **1,000 Clean Tokens** to "power the town" (100% community progress) — encouraging collaboration, not just competition.

### ✓ Progression & Unlocking

Progress is saved on the device; each completed level unlocks the next, so students always have a clear next goal to pursue at their own individual pace.

### 📝 Completion Certificate

Teachers can generate a printable PDF completion certificate for students to celebrate curriculum mastery, persistence, and environmental problem-solving.

## 5 STEM Competencies Tracked

The game tracks a 4-axis STEM competency profile per student, displayed in the classroom portal as a comprehensive radar chart:

### Computational Thinking

Algorithms, sequences, loops, conditionals, and logical problem decomposition.

`CSTA 1B-AP-08, 1B-AP-10`

### Systemic Engineering

Decomposition, structural subsystems, and hardware/software models.

`CSTA 1B-AP-11, 1B-CS-02`

### Mathematical Modeling

Variables, rate calculations, and cause-and-effect input/output relationship models.

`CSTA 1B-AP-09, 1B-DA-07`

### Persistence & Debugging

Iterative code modification, systematic bug isolation, and resilient retry behavior.

`CSTA 1B-AP-13, 1B-AP-15`

**ACHIEVEMENT THRESHOLD**

A competency counts as "achieved" when the average score reaches **0.6 (60%)** or higher across level attempts.

**CURRICULUM ALIGNMENT**

CSTA K-12 Computer Science Standards (Level 1B, ages 8–11) & NGSS Science & Engineering Practices.

## 6 Adaptive Hint System

A 3-tier scaffolded hint system helps students who are stuck, without just giving the answer:

TIER 1

### Observational

Gently points attention to what matters in the current puzzle state (e.g., *"Watch where your bot is facing!"*).

TIER 2

### Conceptual

Explains the underlying logic idea (e.g., *"A Repeat block runs the blocks inside it again and again."*).

TIER 3

### Interactive

Provides an interactive on-screen tool or demo (e.g., loop-unroll visualizer, decision-tree builder, flow-rate simulator).

**Scaffolding Protocol:** Hints appear after a student retries several times (a configured "stuck" threshold), supporting persistence while preserving the joy of independent problem-solving.

## 7 Features for Teachers (Classroom Platform)

Eco-Bot Builders is classroom-ready. The platform has three integrated parts: the **Unity student game**, a **Node.js backend**, and a **React teacher dashboard**. Teacher-facing features include:

### 🔑 Classroom Rooms & Easy Join

Create a room with a 4-digit PIN (and QR code); students join directly from the game using the PIN and their name.

### 📡 Live HUD & Activity Stream

Real-time view of the class: students online, a Live Activity ticker, and the Community Progress bar toward the 1,000 Clean Token goal.

### 🚨 Student Roster & "Needs Help" Alerts

Live status badges show who is progressing smoothly and who is stuck, enabling timely teacher interventions.

### 📊 STEM Competency Radar

Class-average 4-axis radar chart displaying group strengths across Computational Thinking, Modeling, and Persistence.

### 🎮 Classroom Controls

One-click teacher overrides: Pause Class / Eyes Up, Mute Audio, Refresh Roster, and View Analytics.

### 📄 Exports & Certified Reports

Download CSV report for gradebooks and Generate printable PDF completion certificates for student portfolios.

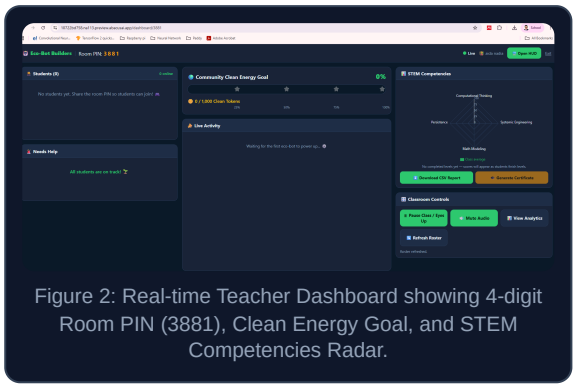



Figure 2: Real-time Teacher Dashboard showing 4-digit Room PIN (3881), Clean Energy Goal, and STEM Competencies Radar.

## 8 Technical Snapshot

**STUDENT GAME CLIENT**

**Unity 2022.3 LTS** (Universal Render Pipeline, URP), written in C#. Cross-platform compatibility covering Windows desktop, Android tablets/emulators, and classroom interactive screens.

**BACKEND SERVICE**

**Node.js & Express** server with **Socket.io** for real-time telemetry streaming and **MongoDB** schema validation for session rosters, telemetry events, and progress metrics.

**TEACHER DASHBOARD**

**React 18** web application with **Recharts** for responsive competency radar charts, **jsPDF** for instant certificate generation, and **socket.io-client** for low-latency live sync.

**Standalone & Networked Play:** Levels 1 and 2 ship as offline, self-contained Unity gameplay (no internet or PIN required to play); the classroom/telemetry platform is for teacher-run sessions. Source code and documentation live in the project's GitHub repository.

STEM-APS-UniMAP